TASK 5: UI DESIGN AND IMPLEMENTATION

Design and Implementation of a Crowdsensed Drive Test Mobile Application

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1. Introduction

This document presents Task 5 of the Crowdsensed Drive Test Mobile Application project, UI Design and Implementation. Building on the system models defined in Task 4, this task covers three interconnected deliverables: the application identity (branding, visual language, and design tokens), the visual design (screen layouts, component library, and interaction patterns), and the frontend implementation (coded screens for both the mobile collector application and the web-based analytics dashboard).

UI design in a professional mobile application is not merely aesthetic, it is a functional requirement. The drive test context imposes extreme usability constraints: the primary user operates the mobile app in a moving vehicle, potentially in direct sunlight, with one hand, while monitoring live signal data. Every design decision must be evaluated against these operational realities. This document demonstrates how design choices directly map to the non-functional requirements (NFRs) defined in the SRS.

2. App Identity

2.1 Brand Name and Concept

The application is named DriveTest Pro. The name was selected to communicate three properties simultaneously: DriveTest establishes the professional domain context immediately, field engineers and RF planners will recognise the term without explanation. Pro signals that this is not a consumer application but a professional-grade tool, setting appropriate expectations for measurement accuracy and workflow sophistication. The combined name is short, memorable, and domain-specific, critical for a tool used in high-cognitive-load field environments.

2.2 Mission Statement

The application's design mission is: Empower field engineers to collect, visualise, and transmit real-time cellular network measurements from everyday smartphones, making professional drive testing accessible, scalable, and affordable.

This mission statement drives three concrete design constraints: accessibility (the UI must be learnable without training for engineers familiar with TEMS Pocket conventions), scalability (the design must work equally well for a single device and a fleet of 50 concurrent devices), and affordability (the app must function on mid-range Android devices, not just flagship hardware).

2.3 Colour Palette

Colour	Hex	Role	Rationale
Navy	#0D1B2A	Primary background (dark theme)	Deep navy reduces OLED power consumption; minimises glare reflection in vehicle windscreens; provides high contrast for all text
Cyan	#00BCD4	Primary accent ; interactive elements, CTAs, active states	Cyan provides maximum contrast against navy (contrast ratio 8.4:1, exceeds WCAG AA 4.5:1 requirement). Associated with technology and signal in telecommunications UI conventions
Signal green	#00E676	Good signal indicator (RSRP > −90 dBm)	Green universally encodes 'good' / 'go'. Consistent with TEMS Pocket and G-NetTrack colour conventions, reduces relearning for professional users
Amber	#FFB300	Fair signal indicator (−90 to −105 dBm)	Amber universally encodes 'caution'. High luminance ensures visibility in sunlight without saturating the display
Alert red	#F44336	Poor signal / coverage hole indicator (< −105 dBm)	Red universally encodes 'danger' / 'stop'. Distinguishable from amber by users with red-green colour blindness (different luminance)

Colour	Hex	Role	Rationale
White	#FFFFFF	Primary text / UI elements on dark backgrounds	Maximum contrast against navy background. Used for primary labels and metric values only never as a background
Slate	#94A3B8	Secondary text / labels / captions	Reduced contrast for secondary information ensures visual hierarchy between primary metrics and contextual labels

2.4 Typography

Role	Typeface	Weight	Size	Usage
Screen titles	Inter (Calibri fallback)	Bold	24px / 18pt	Screen names, section headers, primary labels
Body / labels	Inter (Calibri fallback)	Regular	14px / 10.5pt	Descriptive labels, captions, secondary information
Metric values	Roboto Mono (Courier fallback)	Bold	16–28px / 12–21pt	RF measurement values monospaced for numerical alignment across updates
Dashboard body	Inter (Calibri fallback)	Regular	13px / 10pt	Dashboard text, table content, filter labels

2.5 Iconography

Icon	Usage	Semantic meaning
Concentric circles (logo)	App icon, splash screen	Signal propagation / RF measurement
Filled circle (status dot)	GPS and network status indicators	Green = fixed, Amber = searching, Red = lost / error
Exclamation in circle	Coverage hole markers on map	Alert / warning : coverage gap detected
Signal bars	Network strength quick indicator	4-bar system: 4=excellent, 3=good, 2=fair, 1=poor
Map pin	GPS position indicator	Current device location
Gear	Settings navigation	Configuration access
Download arrow	Export / CSV download	Data export function

3. Visual Design

3.1 Screen Inventory

Screen	Platform	Primary user	Status
Session setup	Mobile (Android/iOS)	Drive test technician	Designed and implemented
Live metrics	Mobile (Android/iOS)	Drive test technician	Designed and implemented
Offline buffer status	Mobile (Android/iOS)	Drive test technician	Designed
Transmission log	Mobile (Android/iOS)	Drive test technician	Designed
Settings	Mobile (Android/iOS)	Drive test technician	Designed
Coverage heatmap	Web dashboard	RF planner, NOC	Designed and implemented
Session list	Web dashboard	RF planner, NOC	Designed
Coverage hole list	Web dashboard	RF planner, NOC	Designed
Session export	Web dashboard	RF planner	Designed
KPI summary	Web dashboard	Executive	Designed

3.2 Mobile App: Session Setup Screen

The session setup screen is the first screen the technician sees when opening the application. Its primary function is to confirm that the device is ready for a drive test: GPS fixed, network connected and to allow the technician to select the session type before pressing start.

3.2.1 Layout structure

- Status bar: system time and battery indicator (24px height)
- App header bar: logo, app name, settings shortcut (42px height)
- Screen title : 'New Session' in bold 24px
- GPS status card: live GPS fix status with coordinates and accuracy radius
- Network status card: current RAT, operator, PCI, and EARFCN
- Session type selector: 4 pill buttons (RF+GPS, Voice, Data, Full). Full pre-selected
- Primary CTA: 'START DRIVE TEST' button in high-contrast cyan, full width
- Bottom navigation: 4 tab icons (Home, Signal, Map, Analytics)

3.2.2 Design decisions

Decision	Rationale	NFR addressed
GPS-first gate session cannot start without GPS	Prevents collection of unlocatable samples that would corrupt coverage maps	FR-02

Decision	Rationale	NFR addressed
fix		
Network status card pre-session	Builds technician trust in data quality before session starts; prevents post-hoc discovery of data quality issues	FR-01
'Full' session type pre-selected	The most complete measurement mode should require zero configuration reduces session start friction	NFR-08
Full-width CTA button	Maximum touch target area (full screen width × 52px height) for reliable tap in a moving vehicle	NFR-08
Bottom navigation tab bar	All primary navigation within thumb reach on 5 -7 inch screens in portrait orientation	NFR-08
Dark theme	Reduces reflective glare in outdoor/vehicle contexts. Consistent with dark cockpit design principles from aviation HMI	NFR-08

3.3 Mobile App: Live Metrics Screen

The live metrics screen is the primary screen during an active drive test session. It is the most glance-intensive screen in the application; the technician checks it briefly, repeatedly, while driving, without sustained focus. Every design decision prioritises information density and readability over aesthetics.

3.3.1 Layout structure

- Live status bar: LIVE indicator, session timer, time of day
- Network context bar: RAT type, operator, PCI, signal bar group
- RSRP primary card: large (28px bold) colour-coded RSRP value with Good/Fair/Poor/Hole label
- Secondary metrics grid: 2×2 compact grid: RSRQ, SINR, DL throughput, UL throughput
- GPS strip: coordinates, speed, and accuracy always visible
- Stop/Upload CTA: full-width red button

3.3.2 Information hierarchy rationale

RSRP is the single most important metric in a drive test, it is the primary coverage indicator and the basis for coverage hole detection. Accordingly, it occupies the top position on the screen, uses the largest font (28px bold), and is colour-coded to the signal quality thresholds. A technician looking at the screen for 0.3 seconds should be able to determine signal quality category from colour alone, without reading the number.

The 2×2 secondary grid uses a uniform 16px bold typeface smaller than RSRP but larger than labels. Each secondary metric card has a coloured border that matches the signal quality of that metric's value, providing a second colour-coding layer without requiring the technician to read individual values.

3.4 Analytics Dashboard: Coverage Heatmap View

The analytics dashboard is a web application consumed by RF planners, NOC analysts, and executives after drive test sessions. Unlike the mobile app, dashboard users are seated at a workstation with sustained attention, the design priorities shift from glanceability to analytical depth and data exploration.

3.4.1 Layout structure

Panel	Width	Content
Top navigation bar	Full width	App logo, nav tabs (Coverage Map, Sessions, Reports, Alerts), Export button
Left sidebar	190px	Filter controls: date range, network type, hole threshold; signal legend
Map area	Central, flexible	Interactive Leaflet/OpenStreetMap map with RSRP heatmap overlay and coverage hole markers
Right KPI panel	200px	Session KPI cards: sample count, avg RSRP, hole count, avg DL, session time

3.4.2 Heatmap design

The coverage heatmap uses a continuous colour gradient mapped to RSRP value ranges: green (> -90 dBm) through cyan (−90 to −95 dBm) through amber (-95 to -105 dBm) to red (-105 to -110 dBm) with coverage holes marked separately as red circles with exclamation markers. The interpolation between sample points uses Kriging (or IDW as an alternative) to fill the gaps between measured locations, producing a continuous surface rather than isolated dots.

Coverage hole markers are visually distinct from the heatmap colouring, they use a bold red circle with an exclamation icon overlaid on the map, making them immediately identifiable even when the surrounding heatmap area is also red. This deliberate visual distinction prevents confusion between 'poor signal' and 'coverage hole' which have different engineering responses.

3.5 Component Library

3.5.1 Metric card component

State	Border colour	Background	Value colour	Use case
Good	#00E676 (green)	#0A2015 (dark green tint)	#00E676	RSRP > −90 dBm
Fair	#FFB300 (amber)	#1A1500 (dark amber tint)	#FFB300	RSRP −90 to −105 dBm
Poor	#F44336 (red)	#200A0A (dark red tint)	#F44336	RSRP −105 to −110 dBm
Hole	#C44B4B (dark red)	#200A0A	#C44B4B	RSRP < −110 dBm coverage hole
Loading	#334455 (slate)	#162032	#94A3B8 (gray)	Waiting for first measurement

3.5.2 Button components

Button type	Fill	Text colour	Usage
Primary / Start	Cyan #00BCD4	Navy #0D1B2A	Starting a session; primary confirmation actions
Destructive / Stop	Red #F44336	White #FFFFFF	Stopping a session; irreversible actions
Secondary / Upload	Blue #1565C0	White #FFFFFF	Upload trigger; secondary actions
Disabled	Slate #334455	Gray #94A3B8	Any action not currently available
Export (dashboard)	Cyan #00BCD4	Navy #0D1B2A	Data export from dashboard

3.5.3 Status indicator system

Status	Dot colour	Label	Context
GPS Fixed	Green #00E676	GPS Fixed · ±Xm	GPS has a satellite fix with accuracy reading
GPS Searching	Amber #FFB300	GPS Searching…	GPS is acquiring satellites; session blocked
GPS Lost	Red #F44336	GPS Lost	GPS signal lost during session ; samples flagged
Network active	Cyan #00BCD4	[RAT] [Operator] Active	Device connected to cellular network
Offline buffering	Amber #FFB300	BufferingX samples queued	No upload connectivity; samples being buffered

Status	Dot colour	Label	Context
Uploading	Cyan #00BCD4	Uploading X/Y samples…	Active upload in progress

4. Frontend Implementation

4.1 Technology Stack

Layer	Technology	Rationale
Mobile app - Android	Kotlin + Jetpack Compose	Native Android; Compose provides declarative UI with reactive state management ideal for live-updating metric displays
Mobile app - iOS	Swift + SwiftUI	Native iOS; SwiftUI provides equivalent declarative UI paradigm to Compose single mental model for both platforms
Dashboard frontend	React 18 + TypeScript	Component-based SPA architecture; TypeScript provides compile-time type safety for data structures from the backend API
Map layer	Leaflet.js + OpenStreetMap	Open-source, no API key required, supports custom tile layers and real-time marker updates
Data visualisation	Recharts (React)	Declarative chart library compatible with React's rendering model; supports real-time data updates
State management	React Context + Zustand	Lightweight; appropriate for dashboard complexity without Redux overhead
API client	Axios + React Query	Type-safe HTTP client; React Query handles caching, background refetch, and loading states
Build tool	Vite	Fast development server and build pipeline; significantly faster than Create React App for development iteration

4.2 Mobile App Implementation: Key Components

4.2.1 Session management

The session manager is the central coordinator of the mobile app. It maintains session state (INACTIVE, STARTING, ACTIVE, STOPPING, UPLOADING) and orchestrates the measurement collection loop, buffer management, and upload trigger. On Android, it runs as a Foreground Service to ensure continuous operation when the screen is locked or the app is backgrounded, a critical requirement for 90-minute drive test sessions.

4.2.2 Measurement collector

The measurement collector polls the Android TelephonyManager at 1Hz using a Handler with a 1000ms delay. On each poll it calls getAllCellInfo() to retrieve the serving cell's RSRP, RSRQ, SINR, PCI, EARFCN, MCC, MNC, and RAT type. The raw CellInfo objects are parsed into the Measurement data class and passed to the GPS stamper.

4.2.3 GPS stamper

The GPS stamper attaches location context to each measurement using the FusedLocationProviderClient (Android) or CLLocationManager (iOS). It requests updates at the highest available accuracy. Each measurement receives the most recent GPS fix, if the fix is older than 5 seconds or has accuracy worse than 50 metres, the sample is flagged with a quality indicator rather than discarded.

4.2.4 Local buffer

The local buffer is implemented as a Room database (Android) or Core Data store (iOS). Each measurement sample is persisted as a row in the measurements table within milliseconds of collection. The buffer manager monitors network connectivity via ConnectivityManager and triggers a flush operation (batch upload) when a suitable connection becomes available. After confirmed upload, flushed samples are marked as uploaded but retained for 24 hours before deletion.

4.3 Dashboard Implementation: Key Components

4.3.1 Coverage heatmap

The coverage heatmap is implemented using Leaflet.js with a custom heatmap layer built on the Leaflet.heat plugin. Each measurement sample is plotted as a weighted point, the weight corresponding to the RSRP value normalised to the 0 - 1 range. The heatmap renders in real time as new session data is received via WebSocket from the backend.

4.3.2 Coverage hole detection overlay

Coverage holes are rendered as a separate Leaflet layer above the heatmap. Each hole location is represented by a custom DivIcon, a red circle with an exclamation mark positioned at the centroid of the detected hole zone. Clicking a hole marker opens a popup with the GPS coordinates, minimum RSRP value observed, number of samples in the hole zone, and timestamp of first detection.

4.3.3 Real-time KPI panel

The KPI panel uses React Query to poll the backend API every 5 seconds for updated session statistics. KPI values animate on update using a CSS counter animation, a subtle effect that draws attention to changed values without disrupting the analyst's focus on the map. If a KPI value worsens (e.g., more holes detected), the value briefly highlights in red before settling to its normal colour.

4.4 Responsive Design

Breakpoint	Target device	Layout adaptation
< 480px (mobile)	Field engineer smartphone	Single column; bottom nav; large touch targets; metric cards stack vertically
480 - 768px (tablet)	Tablet / large phone	2-column metric grid; side drawer replaces bottom nav

Breakpoint	Target device	Layout adaptation
	landscape	
768 - 1280px (laptop)	Dashboard on laptop	Full sidebar + map + KPI panel layout; reduced padding
> 1280px (desktop)	Dashboard on wide monitor	Expanded map area; sidebar and KPI panel at full width; additional detail panels available

4.5 NFR Compliance Summary

NFR	Requirement	Implementation approach	Status
NFR-08	One-handed operation on 5 - 7” screens	Bottom navigation, large CTAs, thumb-reachable controls; minimum touch target 44×44px	Met
NFR-08	Sunlight readability (WCAG AA contrast ≥4.5:1)	Cyan on navy = 8.4:1; white on navy = 18.1:1; all text/background pairs verified	Met
NFR-03	Battery ≤15%/hour during active session	No animations during session; minimal background processing; Foreground Service only when active	Targeted
NFR-04	App size ≤50 MB	Asset optimisation; no bundled map tiles; lazy loading of non-critical components	Targeted
NFR-01	1Hz measurement refresh	Handler-based 1000ms polling loop; UI updates via reactive state (StateFlow / State)	Met
NFR-07	AES-256 at rest for buffer	Room database encrypted using SQLCipher; key stored in Android Keystore / iOS Secure Enclave	Met

References

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